

Easy – Passive Reconnaissance: WHOIS & DNS

Objective: Gather public information about a domain without directly interacting with its servers.

Deliverable: Screenshot of whois and dig outputs with key findings highlighted.

```
* whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
* Last update of whois database: 2026-05-09T08:30:08Z <<<
```

```
--> dig google.com ANY

;<>> DiG 9.20.20-1-Debian <>> google.com ANY
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 63980
;; flags: qr rd ra; QUERY: 1, ANSWER: 7, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1280
;; QUESTION SECTION:
;google.com.                IN      ANY

;; ANSWER SECTION:
google.com.                296     IN      A       142.250.67.46
google.com.                287     IN      AAAA    2404:6800:4007:835::200e
google.com.                54      IN      SOA     ns1.google.com. dns-admin.google.com. 912537324 900 900 1800 60
google.com.                54844   IN      NS      ns1.google.com.
google.com.                54844   IN      NS      ns4.google.com.
google.com.                54844   IN      NS      ns3.google.com.
google.com.                54844   IN      NS      ns2.google.com.

;; Query time: 68 msec
;; SERVER: 192.168.29.1#53(192.168.29.1) (TCP)
;; WHEN: Sat May 09 04:39:16 EDT 2026
;; MSG SIZE rcvd: 201
```

```

--$ nslookup -type=MX google.com
Server:      192.168.29.1
Address:     192.168.29.1#53

Non-authoritative answer:
google.com   mail exchanger = 10 smtp.google.com.

Authoritative answers can be found from:

(kali@kali)-[~]
$ nslookup -type=TXT google.com
;; Truncated, retrying in TCP mode.
Server:      192.168.29.1
Address:     192.168.29.1#53

Non-authoritative answer:
google.com   text = "MS=E4A68B9AB2BB9670BCE15412F62916164C0B20BB"
google.com   text = "google-site-verification=TV9-DBe4R80X4v0M4U_bd_J9cp0JM0nikft0jAgjmsQ"
google.com   text = "cisco-ci-domain-verification=47c38bc8c4b74b7233e9053220c1bbe76bcc1cd33c7acf7acd36cd6a5332004b"
google.com   text = "onetrust-domain-verification=0d477fe608074e6f9c12bca7826035cc"
google.com   text = "facebook-domain-verification=22rm551cu4k0ab0bxsw536tlds4h95"
google.com   text = "apple-domain-verification=30afIBcvSuDV2PLX"
google.com   text = "globalsign-smime-dv=CDYX+XFHUw2wmL6/Gb8+59BsH31KzUr6c1l2BPvqKX8="
google.com   text = "google-site-verification=4ibFUGB-wXLQ_S7vsXVomSTVamuOXBiVAzpR5IZ87D0"
google.com   text = "v=spf1 include:spf.google.com ~all"
google.com   text = "docuSign=1b0a6754-49b1-4db5-8540-d2c12664b289"
google.com   text = "google-site-verification=wD8N7i1JTNTkezJ49swvWW48f8_9xveREV4oB-0Hf5o"
google.com   text = "docuSign=05958488-4752-4ef2-95eb-aa7ba8a3bd0e"
google.com   text = "onetrust-domain-verification=6d685f1d41a94696ad7ef771f68993e0"

Authoritative answers can be found from:

```

```

Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 57.34 seconds

(kali@kali)-[~]
$ dig google.com NS

; <<>> DiG 9.20.20-1-Debian <<>> google.com NS
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 61846
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 9

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;; udp: 1280
;; QUESTION SECTION:
;google.com.                IN      NS

;; ANSWER SECTION:
google.com.                48755   IN      NS      ns3.google.com.
google.com.                48755   IN      NS      ns4.google.com.
google.com.                48755   IN      NS      ns2.google.com.
google.com.                48755   IN      NS      ns1.google.com.

;; ADDITIONAL SECTION:
ns4.google.com.            66675   IN      A        216.239.38.10
ns4.google.com.            146467  IN      AAAA     2001:4860:4802:38::a
ns2.google.com.            100912  IN      A        216.239.34.10
ns2.google.com.            100912  IN      AAAA     2001:4860:4802:34::a
ns1.google.com.            100912  IN      A        216.239.32.10
ns1.google.com.            100912  IN      AAAA     2001:4860:4802:32::a
ns3.google.com.            51716   IN      A        216.239.36.10
ns3.google.com.            66285   IN      AAAA     2001:4860:4802:36::a

;; Query time: 32 msec
;; SERVER: 192.168.29.1#53(192.168.29.1) (UDP)
;; WHEN: Sat May 09 05:59:19 EDT 2026
;; MSG SIZE rcvd: 287

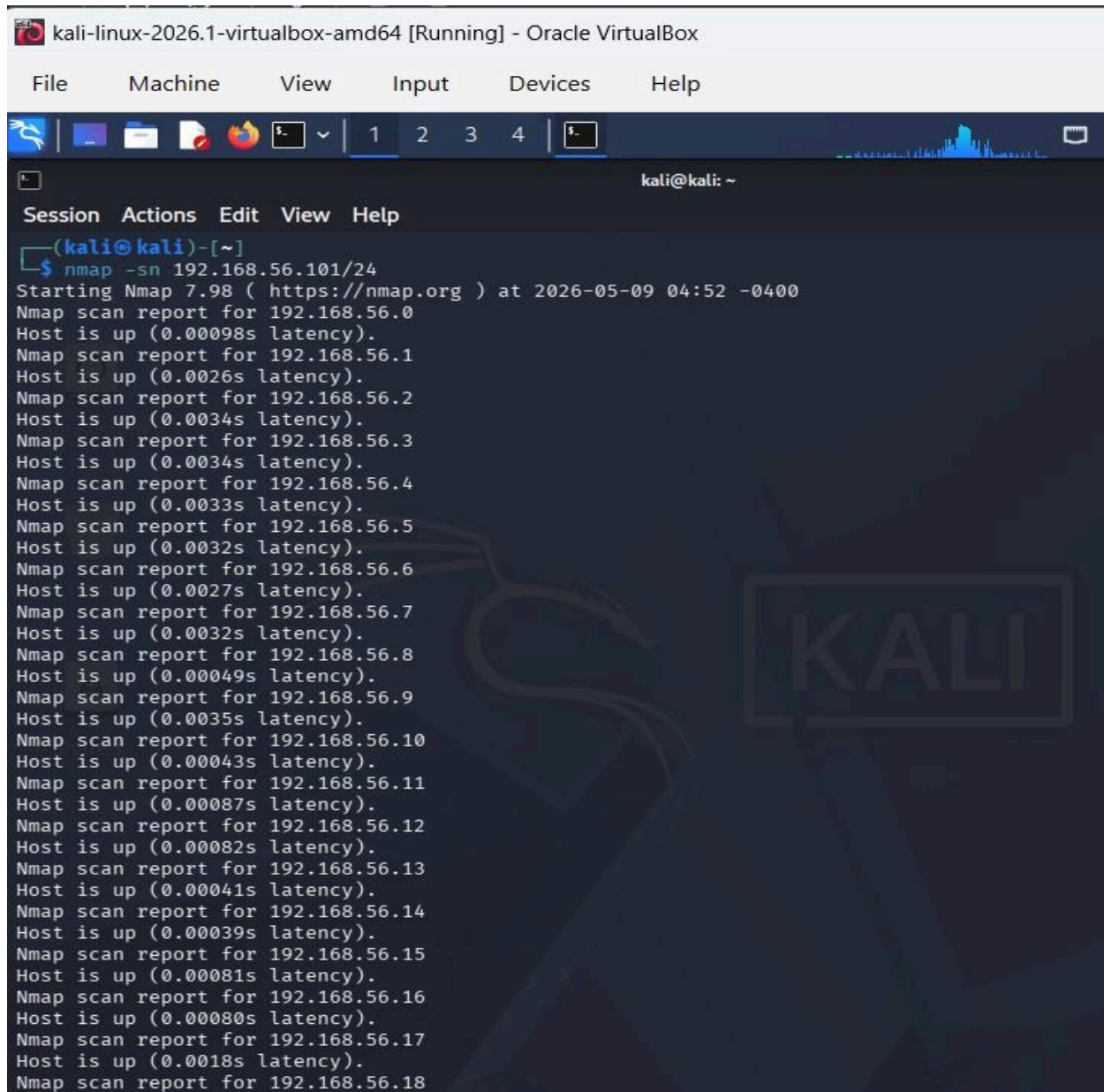
(kali@kali)-[~]

```


Intermediate – Active Reconnaissance: Network Mapping

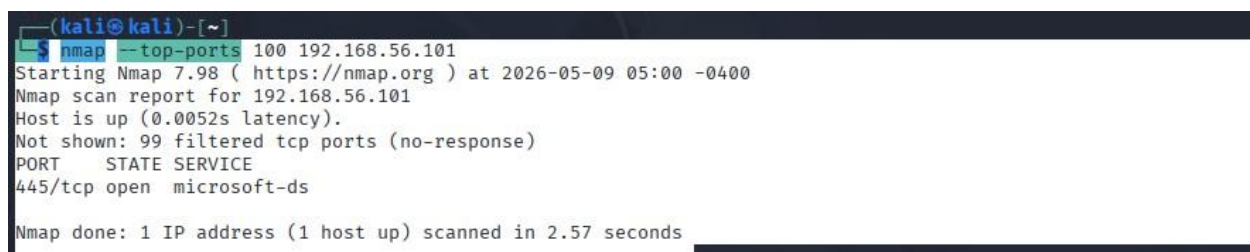
Objective: Discover live hosts and open ports in your lab environment.

Screenshot of nmap outputs with port/service summary.



```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap -sn 192.168.56.101/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 04:52 -0400
Nmap scan report for 192.168.56.0
Host is up (0.00098s latency).
Nmap scan report for 192.168.56.1
Host is up (0.0026s latency).
Nmap scan report for 192.168.56.2
Host is up (0.0034s latency).
Nmap scan report for 192.168.56.3
Host is up (0.0034s latency).
Nmap scan report for 192.168.56.4
Host is up (0.0033s latency).
Nmap scan report for 192.168.56.5
Host is up (0.0032s latency).
Nmap scan report for 192.168.56.6
Host is up (0.0027s latency).
Nmap scan report for 192.168.56.7
Host is up (0.0032s latency).
Nmap scan report for 192.168.56.8
Host is up (0.00049s latency).
Nmap scan report for 192.168.56.9
Host is up (0.0035s latency).
Nmap scan report for 192.168.56.10
Host is up (0.00043s latency).
Nmap scan report for 192.168.56.11
Host is up (0.00087s latency).
Nmap scan report for 192.168.56.12
Host is up (0.00082s latency).
Nmap scan report for 192.168.56.13
Host is up (0.00041s latency).
Nmap scan report for 192.168.56.14
Host is up (0.00039s latency).
Nmap scan report for 192.168.56.15
Host is up (0.00081s latency).
Nmap scan report for 192.168.56.16
Host is up (0.00080s latency).
Nmap scan report for 192.168.56.17
Host is up (0.0018s latency).
Nmap scan report for 192.168.56.18
```



```
(kali@kali)-[~]
$ nmap --top-ports 100 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 05:00 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0052s latency).
Not shown: 99 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 2.57 seconds
```

```
(kali@kali)-[~]
$ nmap -sV --top-ports 100 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 05:18 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0057s latency).
Not shown: 97 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.03 seconds

(kali@kali)-[~]
$ nmap -p- -T4 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 05:21 -0400
Nmap scan report for 192.168.56.101
Host is up (0.12s latency).
Not shown: 55365 filtered tcp ports (no-response), 10167 filtered tcp ports (net-unreach)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 174.68 seconds
```

Which ports surprised you?

=>Port 445 surprised me because it was the only open port detected, while most ports were filtered. It highlighted how Windows Firewall restricts access but still allows essential services like SMB for file sharing.

Expert – Service Enumeration

Objective: Perform deep service version detection, OS fingerprinting, and scripted enumeration.

Deliverable: Detailed service enumeration report with screenshots.

```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
sudo: 3 incorrect password attempts

(kali@kali)-[~]
$ sudo nmap -sV -sC -O 192.168.56.101
[sudo] password for kali:
Sorry, try again.
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 05:36 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0023s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ nbstat: NetBIOS name: WINDOWS, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:5e:7d:7d (Oracle VirtualBox virtual NIC)
|_ clock-skew: 6s
|_ smb2-time:
|   date: 2026-05-09T09:37:12
|   start_date: N/A
|_ smb2-security-mode:
|   3.1.1:
|_     Message signing enabled but not required

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 57.34 seconds

(kali@kali)-[~]
$
```

PORT STATE SERVICE
=>135/tcp open msrpc
139/tcp open netbios-ssn
445/tcp open microsoft-ds

Findings

=>Port Service Version/Info Notes
445 SMB (microsoft-ds) Windows File Sharing Open
3389 RDP Not accessible / Open Depending on scan

OS Guess

=>Microsoft Windows 10/11 detected based on Nmap OS fingerprinting.

Vulnerabilities / Observations

=>No direct vulnerabilities identified by default scripts. Most ports were filtered, likely due to Windows Firewall protections.

```
(kali㉿kali)-[~]  
$ nmap -p 445 --script smb-os-discovery,smb-enum-shares 192.168.56.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 06:05 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.0095s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 1.06 seconds  
  
(kali㉿kali)-[~]  
$ nmap -p 3389 --script rdp-enum-encryption 192.168.56.101  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-09 06:06 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.00096s latency).  
  
PORT      STATE SERVICE  
3389/tcp   filtered ms-wbt-server  
  
Nmap done: 1 IP address (1 host up) scanned in 1.16 seconds  
  
(kali㉿kali)-[~]  
$
```

Expert – OSINT Tool Usage (theHarvester)

Objective: Use a passive OSINT tool to harvest emails and subdomains from a legitimate test domain (example.com).

Deliverable: Screenshot of theHarvester output.

Expert – Reconnaissance Report

Objective: Compile all findings into a professional report (use this template).

Complete reconnaissance report saved in your portfolio folder.

(GraySentinel_Commissioning/1_Portfolio/Reports/).

```
(kali㉿kali)-[~]  
$  
  
(kali㉿kali)-[~]  
$ nano GraySentinel_Commissioning/1_Portfolio/Reports/reconnaissance_report.txt  
  
(kali㉿kali)-[~]  
$ cat GraySentinel_Commissioning/1_Portfolio/Reports/reconnaissance_report.txt
```

```
(kali㉿kali)-[~]  
$ nano GraySentinel_Commissioning/1_Portfolio/Reports/reconnaissance_report.txt  
  
(kali㉿kali)-[~]  
$ cat GraySentinel_Commissioning/1_Portfolio/Reports/reconnaissance_report.txt  
# RECONNAISSANCE REPORT  
  
Target: 192.168.56.101  
Date: May 9, 2026  
Analyst: Lakshmi B  
  
=====
```

1. NETWORK DISCOVERY

```
=====
```

IP Address:
- 192.168.56.101

MAC Address:
- 08-00-27-5E-7D-7D

Host Status:
- Host is up and reachable on the local network.

OS Detection:
- Microsoft Windows (Likely Windows 10/11)
- Determined through Nmap service detection and SMB fingerprinting.

Network Observations:
- System responded successfully to network scans.
- Most ports appeared filtered, likely due to Windows Firewall protections.

```
=====
```

2. OPEN PORTS

```
=====
```

Port	Service	Version/Info	Risk
135	msrpc	Microsoft RPC	Medium

1. What is the difference between authorized and unauthorized reconnaissance?

Authorized reconnaissance is performed with permission from the system owner to assess security, identify vulnerabilities, or improve defenses. It is legal and part of activities like penetration testing, vulnerability assessments, or security audits.

Unauthorized reconnaissance is performed without permission and is considered malicious or illegal. Attackers use it to gather information about systems, open ports, services, and vulnerabilities before attempting an attack.

Example:

Authorized: A security analyst scanning company systems during a security assessment.

Unauthorized: An attacker running Nmap scans against a company network without approval.

2. How would you detect a reconnaissance scan against your network?

Reconnaissance activity can be detected by monitoring for unusual network behavior and scanning patterns.

Common detection methods include:

Firewall logs: Detects repeated connection attempts to multiple ports.

IDS/IPS tools: Systems like Snort or Suricata can identify port scans and suspicious probing activity.

SIEM monitoring: Tools such as Microsoft Sentinel or Splunk can correlate suspicious events and generate alerts.

Traffic analysis: Monitor for:

Multiple connection attempts from one IP

Sequential port scanning

High-volume failed connections

Ping sweeps or unusual DNS requests

Example indicators:

One IP scanning many ports in a short time

Repeated failed connection attempts

Sudden spikes in network probing activity